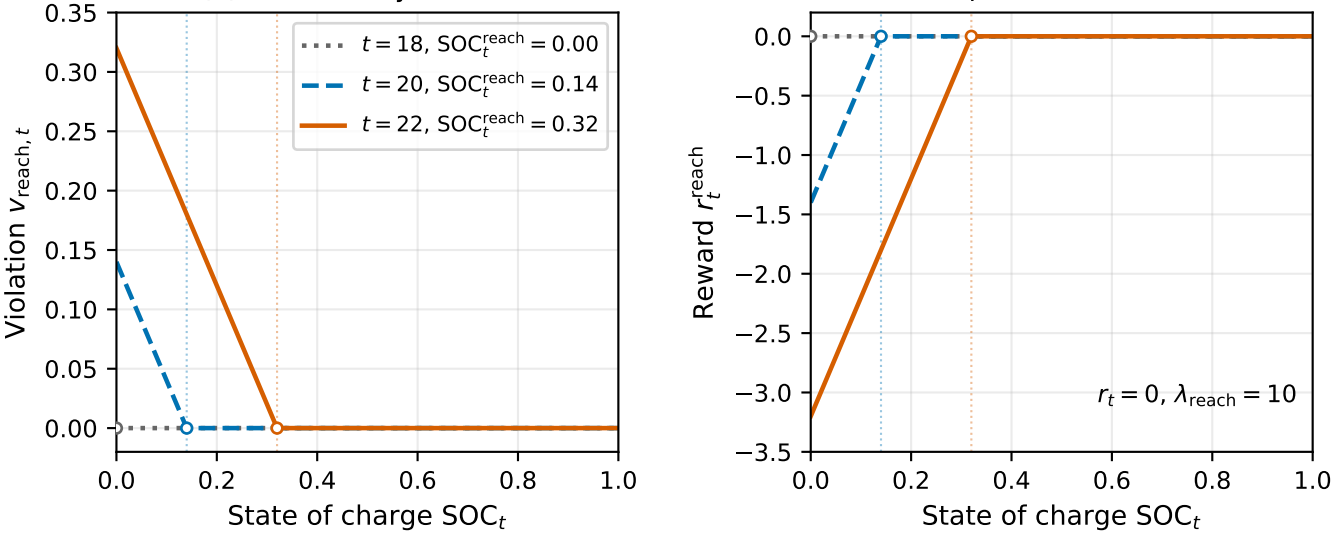
(a) Reachability violation
Violation $v_{\text{reach},t}$
State of charge SOC_t
$t = 18$, $\text{SOC}_t^{\text{reach}} = 0.00$
$t = 20$, $\text{SOC}_t^{\text{reach}} = 0.14$
$t = 22$, $\text{SOC}_t^{\text{reach}} = 0.32$
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.0
0.2
0.4
0.6
0.8
1.0
(b) Shaped reward for $u_t = 0$
Reward r_t^{reach}
State of charge SOC_t
$r_t = 0$, $\lambda_{\text{reach}} = 10$
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5